CONTACT

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RESEARCH INTERESTS

Spatial Intelligence
AI for Sustainability
AI Systems

TECHNICAL SKILLS

Programming: Python, C++, SQL, Java
ML: PyTorch, MindSpore, YOLO, MySQL, Neo4j, MATLAB
Systems: Flask, React, LaTeX

HONORS

National Scholarship
Huawei Scholarship
National Encouragement Scholarship
Tang Zhongying Scholarship
Outstanding Student
Outstanding Student Leader

LANGUAGES

Chinese (native)
English (CET-4 / CET-6)

LEADERSHIP

Tongji University Academic Development and Guidance Association
Head, Jun 2024 – Jun 2025

KUNPENG CHEN

B.Eng. Candidate in Computer Science and Technology
Interested in Spatial Intelligence, Multimodal Generative AI

EDUCATION

Tongji University | Guohao College
B.Eng. in Computer Science and Technology — Sep 2023 – Jun 2027
Selected coursework: Discrete Mathematics, Data Structures, Algorithms, Computer Organization, Database Systems, and Artificial Intelligence.

RESEARCH EXPERIENCE

Cross-Modal Multi-Agent Research
Scientific Intelligence "Hundred Groups, Hundred Projects" — Mar 2026 – Oct 2026
- Conducting ongoing research on cross-modal multi-agent coordination and benchmark design.
- Selected as the only AI4Environment project in Shanghai within the 2025 Scientific Intelligence "Hundred Groups, Hundred Projects" Special Program.

Extending MeCeFO to Additional Architectures
Follow-up study to MeCeFO (NeurIPS 2025) — Mar 2026 – Jun 2026
- Investigating whether MeCeFO-style fault-tolerant optimization remains effective on Mixture-of-Experts (MoE) architectures.
- Studying robustness-efficiency trade-offs under alternative architectures and distributed training conditions.

Trust-Aware Federated Learning for Heterogeneous Edge Systems
Core Member — Mar 2025 – Mar 2026
- Designed a lightweight federated learning framework for heterogeneous edge networks under data drift and poisoning attacks.
- Proposed a trust-topology algorithm based on gradient cosine similarity to isolate anomalous clients and reduce aggregation bias under Non-IID data.

Tianyan Shiju: Vision-based Object Recognition for Building Load Survey
Project Lead — Jan 2024 – Apr 2025
- Led a project combining VR workflows, civil-engineering research, large models, databases, and YOLOv8-based scene object detection for building live-load investigation.
- Developed a high-precision survey method for structural safety assessment and smart-city maintenance, and filed one software copyright application.

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